Context-Aware Fine-Tuning of Large Language Models for First-Innings Score Prediction in T20 Cricket

MSc Viva Presentation

Damith Tilanka Abewardana Ranasingha Ratnayaka

University of Sri Jayewardenepura
Supervisor: Dr. Rajitha M. Silva

June 2026

MSc DSA 055 | Cricsheet T20I data | Llama 3.2-3B + QLoRA

Motivation & Significance

Why T20 score prediction matters

- Broadcast graphics and live commentary
- Fantasy and betting market projections
- Tactical planning for batting and bowling units
- Short format, high variance, rapidly changing innings state

TV | Stats | Market | Teams

Core opportunity

Conventional models usually consume a flat numeric vector:

`overs=12, runs=95, wickets=3, par=165`

But analysts reason in context-rich language:

"95 for 3 after 12 overs, with two strong batters still available, at a 165-par venue."

Research premise: LLMs can exploit this semantic representation after domain-specific fine-tuning.

Research Objectives & Scope

RO1	Construct T20 snapshot dataset with engineered features including RBSS	**RO2**	Design natural-language prompt schema and prompt-completion corpus
RO3	Establish zero-shot Llama 3.2-3B baseline	**RO4**	Fine-tune Llama 3.2-3B using QLoRA SFT
RO5	Compare minimum-validation-loss and final checkpoints	**RO6**	Evaluate GPT-4.1-nano zero-shot as frontier reference

In scope

- International men's T20 matches
- First-innings final score prediction
- Two-over in-game snapshots
- Parameter-efficient fine-tuning

Out of scope

- Domestic leagues such as IPL, BBL, PSL
- Second-innings chase probability
- Full-parameter fine-tuning
- Multi-seed training analysis

Literature Review (1): Themes & Trajectory

Statistical cricket models

DLS/Stern resource framing
Perera–Swartz Markov chains

Strong interpretability, hand-crafted assumptions

Machine learning cricket models

Iyer & Sharda neural nets
Kampakis & Thomas ensembles
Bailey & Clarke wickets-in-hand

Numeric vectors dominate

LLM foundations

Transformer, GPT, Llama
LoRA and QLoRA

Affordable domain adaptation becomes feasible

LLMs for structured prediction

TabLLM, LLM time-series forecasting
Football outcome prediction

Promising, but not yet T20 in-game scoring

Trajectory: hand-crafted resources → numeric ML → language-model structured prediction.

Literature Review (2): Research Gap & Contribution Map

Study / stream	T20 prompt	RBSS-like	Small FT vs frontier	PEFT checkpoint
Perera–Swartz / Markov cricket	No	No	No	No
Kampakis / ML cricket	No	No	No	No
Hegselmann / TabLLM	No	No	No	No
Huckerby / LLM football	No	No	No	No
This thesis	Yes	Yes	Yes	Yes

Synthesised gap

No prior work combines a natural-language T20 match-state prompt, RBSS-style batting-depth encoding, QLoRA fine-tuning of a compact open-source LLM, and a head-to-head benchmark against a frontier zero-shot model.

Methodology Overview

Cricsheet JSON
3,113 matches
→
Filter + venue normalise
2,895 matches
236 venues
→
Feature engineering
16 features
10 snapshots
→
Prompt corpus
28,950 rows
temporal split
→
QLoRA SFT
Llama 3.2-3B
step 800
→
Evaluation + deployment
3 conditions

Data hygiene

No-result and incomplete first innings removed.

Leakage control

Match-level temporal split before evaluation.

Artefacts

HF datasets, HF adapter, W&B logs, Modal service.

Feature Engineering & RBSS

Snapshot construction

- First innings divided into ten 2-over sections
- One prompt-completion pair per section
- Feature groups: state, recent momentum, phase, venue, RBSS
- Venue par score: 52nd percentile of historical first-innings totals

RBSS definition

$$\mathrm{RBSS} = \sum_{p \in \mathcal{R}} w(\mathrm{class}(p))$$

$\mathcal{R}$ = undismissed players at the end of the section.

Player class	**Weight**
Batter	1.00
All-rounder	0.80
Utility player	0.65
Insufficient data	0.45
Bowler	0.30

Worked example

$$1(1.00) + 2(0.80) + 1(0.65) + 3(0.30) = \mathbf{4.15}$$

Why it matters: identical scorelines can have different remaining batting quality.

Prompt Schema & Model Architecture

Prompt skeleton

```
Match type: T20
Venue: {venue}
Venue par score: {par}
Batting team: {team}
Overs completed: {overs}
Balls remaining: {balls}
Runs scored: {runs}
Wickets lost: {wickets}
Current run rate: {rate}
Runs in last 2 overs: {recent}
Dot balls so far: {dots}
Fours hit: {fours} ; Sixes hit: {sixes}
Powerplay completed: {Yes/No}
Remaining Batting Strength Score: {RBSS}
Final 1st innings score:
```

max prompt length: 148 tokens

max seq length: 256

QLoRA fine-tuning recipe

Base model	Llama 3.2-3B decoder-only
Quantisation	4-bit NF4 + double quantisation
LoRA	$r = 16$, $\alpha = 32$, dropout 0.1
Targets	q_proj, k_proj, v_proj, o_proj
Training	1 epoch, lr 2×10^{-4} cosine
Hardware	Kaggle NVIDIA T4, fp16, batch eff. 8

Frozen 4-bit Llama weights → Trainable LoRA adapters → Integer score output

Data, Splits & Tools

Strict temporal split

Split	Date range	Matches	Rows
Train	Up to 2024-12-31	2,408	24,080
Validation	2025-01-01 to 2025-06-30	237	2,370
Test	2025-07-01 onwards	250	2,500

- Match-level chronological partitioning
- Simulates future deployment conditions
- Reduces risk of temporal leakage

Reproducibility stack

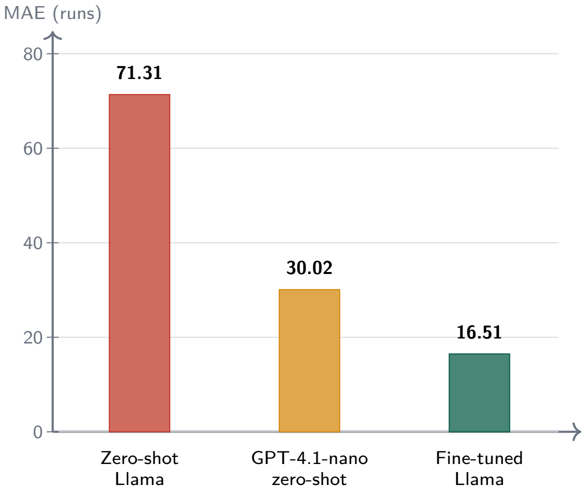
Pydantic v2, SQLite, fuzzywuzzy, Hugging Face Datasets, PEFT, TRL, BitsAndBytes, W&B, LiteLLM, Modal, Gradio, React/Vite, Azure Static Web Apps.

W&B run: v64wpa10

HF adapter: 00be3f3f

Dataset: 28,950 rows

Results: Fine-Tuned 3B Beats Zero-Shot Frontier



Cross-condition metrics

Model	MSE	R^2
Llama ZS	7,463	−317.4%
GPT-4.1-nano ZS	1,538	14.0%
Llama FT	**543**	**68.3%**

vs Llama ZS: 76.8% lower MAE

vs GPT-4.1: 45.0% lower MAE

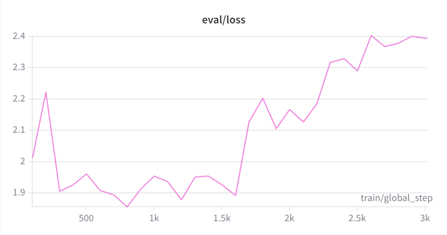
R^2 gain: +54.3 pp vs GPT-4.1

Discussion: Why Fine-Tuning Wins

Interpretation

- General pre-training sees cricket prose, not the conditional mapping $P(\text{final} \mid \text{state})$.
- SFT supplies 24,080 domain examples of that mapping.
- GPT-4.1-nano tends to under-project early states: e.g. 14/1 after 2 overs at par 160 $\rightarrow$ predicted 89, actual 160.
- RBSS and venue par score enrich the prompt beyond raw score and wickets.

Checkpoint selection



Minimum validation loss: **1.856 at step 800**

Checkpoint	N	MAE	R^2
Step 800	500	**16.51**	**68.3%**
Final	500	16.54	68.1%

Limitations

Scope limitations

- International men's T20 only
- Domestic leagues excluded
- First innings only
- Women's matches not modelled in this study

Evaluation boundaries

- GPT-4.1-nano evaluated at $n = 200$ due to API cost

Methodological limitations

- Single fine-tuning run; no multi-seed variance
- 3B base model only; larger open-source models untested
- Validation loss computed on 50-row subsample
- Greedy point prediction only; no uncertainty intervals
- Regex fallback-to-zero inflates zero-shot Llama error

Conclusions & Future Work

Conclusions

1. Domain-specific fine-tuning transforms zero-shot Llama from unusable to competitive.
2. A compact fine-tuned open-source model beats a frontier zero-shot model on this structured numerical task.
3. Minimum-validation-loss checkpoint selection provides a small but consistent generalisation benefit.

Core claim: small, fine-tuned, domain-specific LLMs can be practical sports forecasting tools.

Future work

- Extend to IPL, BBL, PSL and other domestic leagues
- Evaluate 7B, 8B and 70B open-source models
- Build second-innings chase prediction
- Add uncertainty via sampled completions
- Run feature ablations, especially RBSS and venue par
- Integrate live ball-by-ball feeds

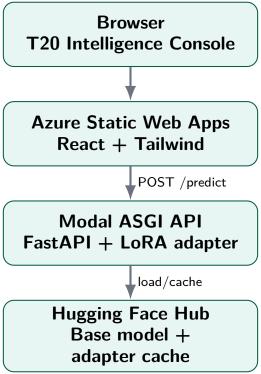
Practical Implications

For cricket analytics teams

- Prioritise domain data curation and PEFT over frontier API prompting
- Host small fine-tuned models on controlled infrastructure
- Use point predictions for graphics and simulation, not high-stakes settlement

Deployment validation

- Modal GPU backend: about 30 s cold, about 2 s warm
- Gradio prototype on Hugging Face Spaces
- React/Vite dashboard on Azure Static Web Apps



`https://mango-ocean-00260de00.7.azurestaticapps.net/`

References (Selected)

1. Vaswani et al. (2017). *Attention Is All You Need.*
2. Brown et al. (2020). *Language Models are Few-Shot Learners.*
3. Touvron et al. (2023). *LLaMA: Open and Efficient Foundation Language Models.*
4. Hu et al. (2022). *LoRA: Low-Rank Adaptation of Large Language Models.*
5. Dettmers et al. (2023). *QLoRA: Efficient Finetuning of Quantized LLMs.*
6. Hegselmann et al. (2023). *TabLLM: Few-shot Classification of Tabular Data with LLMs.*
7. Gruver et al. (2023). *Large Language Models are Zero-Shot Time Series Forecasters.*
8. Huckerby et al. (2024). *Leveraging LLMs for Football Match Outcome Prediction.*
9. Duckworth and Lewis (1998). *A Fair Method for Resetting Targets.*
10. Bailey and Clarke (2004). *Predicting Match Outcome in ODI Cricket.*
11. Cricsheet (2024). *A Cricketing Archive.*

Full bibliography appears in the thesis reference list.

Thank you. Questions?